

Explainable AI for cooling load prediction and heat flow estimation using SHAP values

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Abstract

Effective load management for cooling energy can be realised through accurate cooling load predictions. This can reduce peak loads and ease the burden on the power grid. Accurate load prediction can be achieved using machine learning, which are black-box models. Explainable AI can be used to make machine learning models explainable. SHAP values are employed for this purpose. Currently, there is a research gap as no comprehensive and systematic investigation of the influence of building characteristic on the SHAP values of hourly cooling load predictions has been carried out. Moreover, it has not yet been investigated whether the heat flows in a room can be represented by the local SHAP values. This study investigates the influence of individual features on the cooling load predictions. The feature importance was examined both local, for a single point in time, and global, for the entire dataset, using SHAP values for a large number of various buildings with different characteristics. The results show a correlation between building characteristics and the importance of features for the cooling load prediction. Therefore, the building characteristics are implicit in the SHAP values. This enables an purely data-driven interpretation of the building characteristics. Additionally, the study demonstrated that the local SHAP values provide a satisfactory description of the heat flows in a room. This provides a purely data-driven insight into the heat flows of the building, which can reveal optimisation potential for the building.

Introduction

The increasing demand for cooling energy in newer buildings, driven by rising outdoor temperatures and improved insulation, underscores the need for effective load management strategies. Moreover, the global space cooled in buildings is projected

to increase from $105 \times 10^9 \text{ m}^2$ in 2022 to $250 \times 10^9 \text{ m}^2$ in 2050. The share of space cooling in final electricity consumption is expected to be 9 % in 2022, falling to 5 % in 2050 ([IEA 2023b](#)). Cooling plays a significant role in achieving the energy transition, despite its decreasing share of total electricity consumption. To meet the *Net Zero Roadmap*, the share of solar PV and wind in total global electricity generation will increase from 12 % to 71 % in 2050 ([IEA 2023a](#)).

Integrating load management for cooling systems is crucial due to the fluctuating power generation of these energy sources. The objective is to generate electricity and cooling energy simultaneously to the greatest extent possible. Load shifting of cooling energy is a viable option as thermal energy storage is more cost effective than electrical energy storage ([Odukamaiya et al. \(2021\)](#)). To ensure successful load shifting, accurate prediction of cooling load is required. Using machine learning (ML) to predict the cooling load has the advantage that no information about the building is required and the prediction can be entirely data driven. This means there is no need for a complex modelling of the building, as is the case with grey box modelling. ML models are black boxes, lacking transparency and explainability. Through the use of Explainable Artificial Intelligence (XAI), it is possible to understand how ML models arrive at their results. XAI provides insight by transforming the black box model into a kind of grey box model. This study confidently investigates the use of XAI for explaining cooling load predictions. In this study, SHapley Additive exPlanations (SHAP) values are used as the method for XAI. SHAP values explain how individual features affect a model's output, making the model more explainable.

[Moon et al. \(2022\)](#) investigated the explainability of electrical forecasting in a university building using SHAP values, specifically the Tree SHAP algorithm. Various machine learning models were used for the prediction, including Random Forest

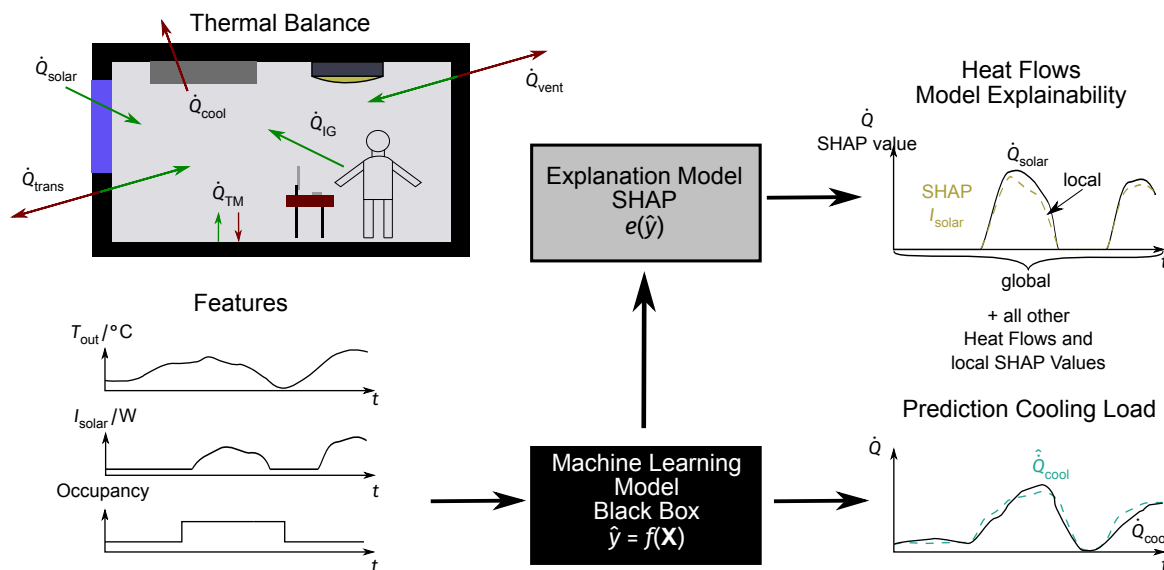


Figure 1: Overview of the work

(RF), Extreme Gradient Boosting (XGBoost) and Light Gradient-Boosting Machine (LightGBM). Among these models, LightGBM achieved the highest prediction quality. The weekly average electricity consumption and the previous day's electricity consumption had the greatest influence on the prediction. In Kuzlu et al. (2020), the XAI models LIME, SHAP and ELI5 were used to investigate which parameters have the greatest influence on solar photovoltaic power generation prediction. The sum of the SHAP values was compared with the prediction for selected time stamps. The influence of each feature could be visualised for single time steps. Neubauer et al. (2024) analysed the influence of individual features on the prediction of the heating load, based on the building characteristics. The study has conclusively shown that the building characteristics are implicitly included in the local and global SHAP values for heating load prediction.

There is still a research gap for investigating the model explainability for the cooling load prediction as a function of the building characteristics. Furthermore, the capability of the local SHAP values to represent the heat flows in a building has not been investigated. Figure 1 presents an overview of the study. The machine learning model, which is a black box model, solely learns the mapping between the features and the prediction. The explanation model, which utilises SHAP values, is used to present an explanation of the relationship between the features and the

prediction. The SHAP values are examined to see whether the building characteristics are represented in the SHAP values and whether the thermal balance and associated heat flows can be represented by the local SHAP values. The objective is to enhance the transparency of the model and obtain insights based solely on data driven analysis of the building. The main contributions of this paper are:

- Predict the cooling load using LR, RF, XGBoost and LightGBM
- Show the relationship between the SHAP value for cooling load as a function of building characteristics
- Comparison of the local model explainability using SHAP values with the heat flows in a room model

Simulation

Machine Learning Regression Models

Several models were evaluated to determine the most appropriate one for this study. Four supervised learning models are considered Linear Regression (LR), RF, XGBoost, and LightGBM. LR is a statistical model, while RF, XGBoost, and LightGBM are decision trees. According to Wang et al. (2020), RF and XGBoost had the highest prediction accuracy for the hourly cooling load prediction of the shallow machine learning models.

In the work of [Moon et al. \(2022\)](#), the LightGBM was able to deliver good results in electrical load prediction. The most appropriate model for this prediction is determined by considering multiple models.

Evaluation Metrics

The evaluation of the model accuracy uses the *root mean squared error* RMSE (Equation 1) and the *coefficient of determination* R^2 (Equation 2).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y})^2} \quad (1)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

The RMSE is the root mean square error between the actual y and the predicted value $\hat{y}$. The unit of the RMSE is equal to the variable y and thus is dimensionally related. In order to compare between different time spans, dimensionless parameters such the R^2 are used. The R^2 can be between 0–1, with higher values indicating a better prediction accuracy.

Explainable AI Models

SHAP values can be used to determine the influence of features on a prediction. They are a game-theoretic approach introduced by [Lundberg & Lee \(2017\)](#). SHAP values are used to justify the output of machine learning models, increasing the transparency, explainability and interpretability of machine learning models. Model Explanations can be created either locally, for a single prediction, or globally, for all predictions in the dataset. The SHAP values are based on the Shapley values ϕ_j (Equation 3), which determine a fair share of the payout for a cooperative game. The Shapley value is calculated by summing the weighted marginal contributions of all possible coalitions. This is done by determining the influence of each player $1, \dots, |N|$ in the coalition of all players N . Every possible coalition of players is S . The size of the coalition is $|S|$, the value function $v(S)$ contains all possible combinations of all players. The payout for all players in the coalition is $v(N)$. The marginal contribution $(v(S \cup \{j\}) - v(S))$ indicates the increase in payout if the player j joins the game. This is the core of the calculation. Therefore, the Shapley value is the sum of all weighted marginal contributions across all possible coalitions without player j .

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \underbrace{\frac{|S|!(N - |S| - 1)!}{N!}}_{\text{weight of marginal contribution}} \cdot \underbrace{(v(S \cup \{j\}) - v(S))}_{\text{marginal contribution}} \quad (3)$$

The Shapley values fulfil the following properties:

1. Efficiency: The sum of all contributions is the payout
2. Symmetry: Two equal players receive the same payout
3. Dummy: If all marginal contributions are zero, the Shapley value is also zero
4. Additivity: For a game with two value functions v_1 and v_2 , the Shapley values for the sum of the games can be expressed as the sum of the Shapley values

The naming convention for SHAP values differs from that of Shapley values. In the case of SHAP values, it is features instead of players and predictions instead of payouts. The calculation of the SHAP value function v is not shown in this paper, a detailed description can be found in [Molnar \(2023\)](#). There are several methods for calculating the SHAP value, including the *Kernel SHAP* algorithm and the *Tree SHAP* algorithm, which are compared in this work.

The Kernel SHAP (KS) is a kernel-based estimation approach for Shapley values, inspired by Local Interpretable Model-agnostic Explanations (LIME). It is a model-agnostic method that estimates SHAP values by taking samples of coalitions and performing a weighted linear regression. The linear model is trained and the estimated coefficients of the model are the SHAP values. The Tree SHAP (TS) method is for tree based models such as RF, XGBoost or LightGBM. It is therefore a model-specific method. One advantage of TS is that they require less computing time compared to KS. In addition, it is an exact method, i.e. you get the correct SHAP values and not just estimates, at least as far as the coalitions are concerned. This work considers only the Interventional Tree Estimator, which uses the tree structure to calculate SHAP values. In this case, the estimation is performed by recursively traversing the tree(s) ([Molnar 2023](#)).

Building Description

The cooling load prediction data set is generated by simulating a simplified dynamic room model. The room is modelled using two RC elements. All exterior and interior components are combined

into one equivalent element each. The area of the room is 17.5 m^2 and the room has a south facing exterior wall with a window. The exact room model is described in [Lauster et al. \(2014\)](#). The room is modelled using the modelling language *Modelica*. The room model used is the thermal zone with two elements for exterior and interior walls model of the Buildings library version 8.0.0 ([Wetter et al. 2014](#)). A full factorial design of the simplified room model, comprising different room surfaces (Table 1) and building characteristics (Table 2), was employed to determine the annual hourly cooling load for 15 552 various room variants. The dataset can be accessed via the [GitHub repository](#).

The outer wall area A_{wall} and the window area A_{win} were varied to generate different room types. The surface properties of the room types are shown in Table 1. The window to wall ratio $A_{\text{win}}/A_{\text{wall}}$ and the wall area to room volume ratio $A_{\text{wall}}/V_{\text{room}}$ are varied by adjusting the A_{wall} and A_{win} . The ratio $A_{\text{win}}/A_{\text{wall}}$ has a major influence on the solar gains and the transmission.

Table 1: Investigated room surfaces

Room type	A_{win} m^2	A_{wall} m^2	$A_{\text{win}}/A_{\text{wall}}$ 1	$A_{\text{wall}}/V_{\text{room}}$ m^{-1}
1	7	3.5	0.67	0.2
2	3.5	7	0.33	0.2
3	7	18.5	0.27	0.49

By varying the building characteristics (see Table 2) and the room surfaces in a full factorial design, the thermal balance in the different room variants changes, in some cases considerably. Due to the wide range of building characteristics, a large number of different building ages and room characteristics can be represented. Figure 1 shows the heat flow in the room, the transmission $\dot{Q}_{\text{trans}}$, the ventilation $\dot{Q}_{\text{vent}}$, the internal gains $\dot{Q}_{\text{IG}}$, the solar gains $\dot{Q}_{\text{solar}}$ and the heat storage and release in the thermal mass $\dot{Q}_{\text{TM}}$. The required cooling load $\dot{Q}_{\text{cool}}$ results from these heat flows.

The internal gains $\dot{Q}_{\text{IG}}$ act as convective and radiant heat in the space. They are simplified to the specific internal gains $\dot{q}_{\text{IG}}$. The air exchange rate (AER) and $\dot{q}_{\text{IG}}$ correspond to the values in Table 2 during the periods of use Monday to Friday from 7 am - 5 pm. Outside of the usage times, AER and $\dot{q}_{\text{IG}}$ are 0. The International Weather for Energy Calculations (IWEC) is used for the German cities of Berlin and Stuttgart, which allows different climate zones to be taken into account.

Table 2: Investigated building characteristics

Nr.	U_{wall} $\text{W/m}^2\text{K}$	U_{win} $\text{W/m}^2\text{K}$	g-value 1	AER h^{-1}	$\dot{q}_{\text{IG}}$ W/m^2
1	0.15	0.8	0.15	0.25	5.1
2	0.25	1.1	0.3	0.5	10.3
3	0.5	1.4	0.45	0.75	15.4
4	0.75	1.7	0.6	1	20.6
5	1	2.1	0.75	2	25.7
6	1.5	2.7	0.9	4	30.9

Results Analysis and Discussion

Thermal Balance

The data set for the cooling load prediction is from 1 June to 15 September (90 days). The training set is from 1 June to 5 August 2023 (72 %) and the test set is from 6 August to 31 August (28 %). Figure 2 shows the cooling energy demand Q_{cooler} for the test set depending on the building characteristics. The cooling energy demand values range from 5.1 to 552.5 kWh with a median of 182.2 kWh. The median values are represented by the olive-coloured line.

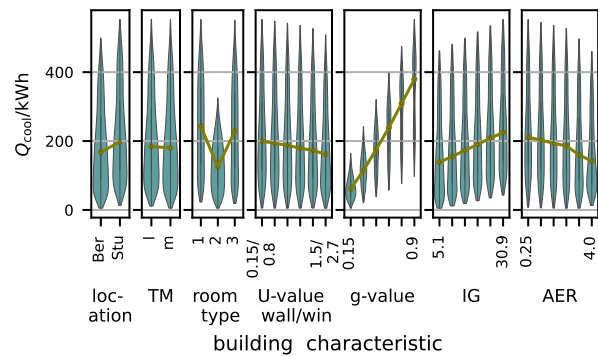


Figure 2: Thermal balance of the various room variants

The combination of a low g-value, low internal gains (IG) and room type 2 (small window area) results in low cooling energy demand values. In terms of solar radiation, the Stuttgart location has a higher $\bar{i}_{\text{sol,so}}$ of 86.8 W m^{-2} compared to Berlin 77.54 W m^{-2} . The $\bar{T}_{\text{out}}$ are similar, with Stuttgart at 18.84°C and Berlin 18.58°C . Due to the higher irradiation, the cooling energy demand Q_{cool} is higher in Stuttgart. The larger window area in room type 1 (7 m^2) compared to room 2 (3.5 m^2) results in a higher Q_{cool} .

An ANOVA is used to determine which building

characteristic has the greatest influence on the cooling load. The F-values are shown in Table 4. All building characteristics have a significant influence on the cooling load ($p < 0.05$). The F-value is the ratio of the variation between sample means to the variation within the samples. The g-value has the largest F-value. Therefore, the variability of the cooling load is most influenced by this factor. The thermal mass has the lowest F-value.

SHAP Values

For the calculation of the SHAP values, we compare the KS and TS algorithm. KS uses a weighted linear regression, while TS uses tree models to determine the SHAP values. In order to select a suitable tree model, the LightGBM, XGBoost and RF variants are compared with each other. The 1h ahead cooling load prediction uses the three features outside temperature T_{out} , solar irradiance on south-facing vertical surfaces $I_{sol,so}$ and occupancy (occu). Only these three features have a relevant feature importance (FI). Lagged or averaged features were not taken into account in this study. A hyperparameter search was conducted in advance for the models, and the most suitable hyperparameters were selected. Table 3 shows the RMSE and R^2 of the different prediction models for the selected standard room: location Berlin, thermal mass m , room type 2, U-values (wall: $0.75 \text{ W m}^{-2} \text{ K}$, window: $1.7 \text{ W m}^{-2} \text{ K}$), g-value 0.45, $\dot{q}_{IG}$ 10.3 W m^{-2} and AER 0.5 h^{-1} .

Table 3: Prediction accuracy for different ML models

Model Type	Train Set		Test Set	
	RMSE/W	R^2	RMSE/W	R^2
LR	114.4	0.7415	129.3	0.7886
LightGBM	51.4	0.9478	89.2	0.8992
XGBoost	53.3	0.9440	100.2	0.8729
RF	53.0	0.9446	95.8	0.8839

The LightGBM model has the lowest RMSE and the highest R^2 for the test and training set. Therefore, its prediction accuracy is the best. The high R^2 value and the low RMSE value indicate a good prediction quality. Figure 5 shows a section of the time course of the prediction and the actual value. The XGBoost and RF models also achieve similarly good values, while linear regression delivers the worst prediction accuracy. Due to its best prediction accuracy, the TS algorithm uses

the LightGBM model.

The LightGBM and TS algorithm were used to calculate the relative proportion of the three SHAP values for each of the 15552 room variants. Figure 3 shows the relative SHAP values as a function of the building characteristics. The median values are represented by the olive-coloured line. For example, higher U-values lead on average to an increasing SHAP value of T_{out} and a decreasing SHAP value of $I_{sol,so}$, whereas an increasing g-value leads on average to a smaller SHAP value of T_{out} and occu and a larger SHAP value of $I_{sol,so}$. The building characteristics are therefore implicitly included in the SHAP values of the cooling load. $I_{sol,so}$ has the highest mean relative SHAP value, followed by T_{out} and occupancy.

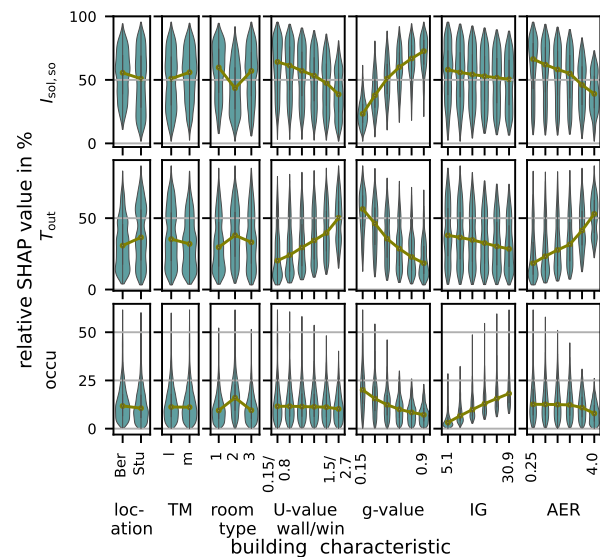


Figure 3: SHAP values depending on the building characteristic

A MANOVA was used to analyse the influence of the building characteristics (independent variables) on the SHAP values (dependent variables). The Wilks Lambda (WL) values of the building characteristics for the individual SHAP values have a similar ranking order as the F-values of the cooling energy demand (refer to Table 4).

A large or small variability of the building characteristics on the cooling energy demand therefore leads to a large or small influence of the relative SHAP values of the features on the cooling load. The building characteristics are therefore implicitly included in the SHAP values. A correlation between global model explainability based on SHAP values and building characteristics has been shown. In addition to

the global consideration (whole dataset), the model explainability can also be performed on a local level. For this purpose, the SHAP values are analysed for each time step.

Table 4: ANOVA and MANOVA of the building characteristics on cooling energy demand and SHAP values

Building Characteristic	Thermal Balance F-value	Rank	SHAP values WL	Rank
location	1152	4	0.8508	3
thermal mass	22	1	0.9280	2
room type	106	2	0.9774	1
U-value	588	3	0.4420	4
g-value	38414	7	0.2707	7
IG	3083	6	0.4072	5
AER	1818	5	0.3460	6

Figure 4 shows the local SHAP values determined with the TS as a function of the feature values for the standard room. The SHAP values increase with the feature values T_{out} , $I_{sol,so}$ and occu. The highest Pearson correlation coefficient r is observed between the occupancy value and the SHAP value of occupancy. There is also a strong correlation between the SHAP value T_{out} and the feature value T_{out} , as well as between the SHAP values $I_{sol,so}$ and the feature values of $I_{sol,so}$. Hence, there is a strong correlation between the three features and the corresponding SHAP values.

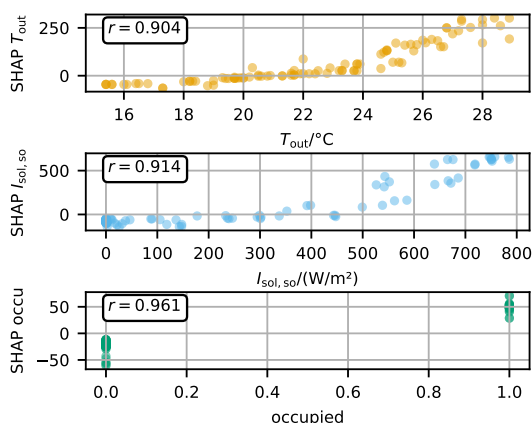


Figure 4: SHAP values depending on the feature values

Figure 5 shows the cooling load $\dot{Q}_{cool}$, local SHAP values and scaled features for two rooms from 6 to 12 August. This makes it possible to visualise the interaction between the individual variables. The LightGBM model with TS algorithm

was used. The first room is the standard room 1. Room 2 has the same characteristics as room 1, but has a higher U-value (wall: $1.5 \text{ W m}^{-2} \text{ K}$, window: $2.7 \text{ W m}^{-2} \text{ K}$), a lower g-value (0.15) and a higher internal occupancy (30.9 W m^{-2}).

The first subplot shows the cooling load and the predicted cooling load $\hat{\dot{Q}}_{cool}$ for rooms 1 and 2. The predicted value of the cooling load from the LightGBM model is equal to the sum of all SHAP values and the base value, which is the mean value of the prediction over the train set. Subplots 2 and 3 show the local SHAP values of the three features (T_{out} , $I_{sol,so}$ and occu). A negative SHAP value will decrease $\hat{\dot{Q}}_{cool}$ and a positive value will increase it. The absolute sum of the SHAP values for the period from 8 to 12 August is shown under the feature names. A higher value means a greater importance of the feature.

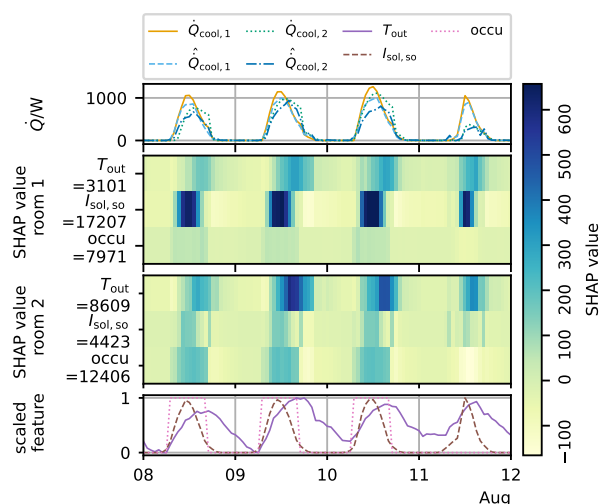


Figure 5: $\dot{Q}_{cool}$ and local SHAP values for two rooms depending on the feature values

The SHAP values are affected by the building characteristics (see Figure 3) and the feature value (subplot 4). Room 1 has a lower U-value, resulting in lower T_{out} SHAP values. The T_{out} SHAP values for room 2 (subplot 3) show a clear difference between the night hours with low T_{out} and the day hours with higher T_{out} values (subplot 4). The g-value of the windows is higher for room 1 at 0.45 than for room 2 at 0.15. This results in significantly higher sums of the absolute SHAP values of $I_{sol,so}$ for room 1. These higher SHAP values can also be seen in the temporal patterns of the local SHAP values of $I_{sol,so}$ (compare subplot 2 and 3). The influence of occupancy can be seen clearly on 11 August. In room 2, with very high internal loads, the cooling load differs significantly between the

days with occupancy (8-10 August) and without occupancy (11 August), while in room 1 the cooling load decreases only slightly on 11 August (subplot 1). The higher differences are also reflected in the SHAP values for occu between room 1 and room 2. There is therefore a correlation between the feature values and the local SHAP values over the course of time.

Comparison SHAP Values with Heat Flows

The following analysis examines whether the local SHAP values can also be used to describe the heat flows in the building. To achieve this, the SHAP values are determined using the TS and the KS algorithm. Figure 6 shows a comparison of the heat flows with the local SHAP values for room 1 during the period from the 8th to the 12th of August (part of the test set). The first subplot shows the cooling load $\dot{Q}_{cool}$ and the predicted cooling load $\hat{\dot{Q}}_{cool}$, which is equal to the sum of all SHAP values and the base value. The second subplot shows the SHAP value of $I_{sol,so}$ and the heat flow of solar radiation $\dot{Q}_{solar}$. The graph of the two variables is similar and the R_2 between $\dot{Q}_{solar}$ and $I_{sol,so}$ value is 0.754 for the TS and 0.602 for the KS.

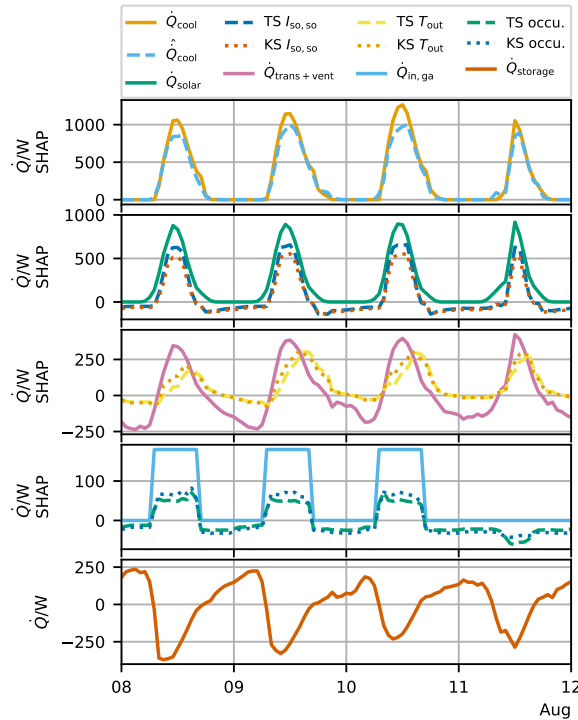


Figure 6: Comparison of SHAP values (Tree SHAP and Kernel SHAP) and the thermal balance

The comparison between the SHAP value of T_{out} and the heat flow of transmission and ventilation $\dot{Q}_{trans+vent}$ (subplot 3) shows a R_2 of 0.444 (TS) and 0.603 (KS). It is important to note that the SHAP values do not represent negative heat flows for $\dot{Q}_{trans+vent}$ at night ($T_{out} < T_{room}$). This is due to the storage and release of energy in the thermal mass $\dot{Q}_{TM}$ (subplot 5), which is not explicitly considered by the selected features and therefore by the SHAP values. This issue is also evident in the internal heat gains $\dot{Q}_{IG}$ (subplot 4), which are poorly represented by the SHAP values of occu, as indicated by the low R_2 values of 0.278 (TS) and 0.169 (KS).

Table 5 presents the R^2 values for the heat flows and the SHAP values for the KS and TS. To investigate the influence of the model prediction accuracy on the prediction accuracy of the SHAP values on the heat flows, the R^2 for the time period (TP) 1 (08-12 Aug) and the TP 2 one week later (15-22 Aug) are examined.

Table 5: Evaluation of the mapping between the heat flows and the courses of the local SHAP values

Energy	SHAP value	Kernel SHAP (KS)		Tree SHAP (TS)	
		TP 1 R^2	TP 2 R^2	TP 1 R^2	TP 2 R^2
$\dot{Q}_{cool}$	$\hat{\dot{Q}}_{cool}$	0.959	0.898	0.959	0.898
$\dot{Q}_{solar}$	I_{sol}	0.602	0.385	0.754	0.598
$\dot{Q}_{trans+vent}$	T_{out}	0.603	0.309	0.444	0.126
$\dot{Q}_{IG}$	occu	0.278	0.210	0.169	0.141

Notably, in TP2, the R^2 between $\hat{\dot{Q}}_{TM}$ and $\dot{Q}_{stor}$ is lower, resulting in significantly lower R^2 between the heat flows and local SHAP values. Therefore, a good description of the heat flows by the local SHAP values is only achieved if the prediction quality of the ML model reaches good values. The KS has a higher R^2 than the TS in two out of three cases for TP 1 and TP 2. A more accurate description of the heat flows is therefore achieved with the KS, which estimates the SHAP values by using a special weighted linear regression. The calculation time for the KS is 20.960 s, many times longer than that of the TS, which is only 0.012 s.

In general, it can be stated that local SHAP values provide a limited representation of the heat flows. A more accurate representation would require information about the thermal mass properties. A feature that provides information on the temperature of the thermal mass is only realistic in simulations and is therefore

unsuitable. However, it is possible to represent the thermal behaviour of the building using lagged features or sequence-to-sequence models. These models take into account a number of past features to provide multiple step ahead predictions. Furthermore, the validation of the presented method on real buildings is challenging due to the unavailability of heat flow information and the difficulty in determining it. In addition, the simplified occupancy patterns and the consideration of only one room result in a significant simplification. Nevertheless, it has been shown that the local SHAP values can predict the heat flows of a room with satisfactory accuracy using a purely data-driven approach. This provides additional information and insights that can be used to identify optimisation potential for the building.

Summary

Accurate load prediction is crucial for cooling load management, and this can be achieved using ML models, which are black-box models. By using XAI, these models can be made explainable. In this study, SHAP values were used to analyse XAI for the cooling load of 15552 different simulated room variants. Accurate load prediction is crucial for cooling load management, and this can be achieved using ML models, which are black-box models. The SHAP values implicitly contain information about the building characteristics, and the g-values have the greatest discriminatory power on the SHAP values. The influence of features on the room the local SHAP values can satisfactorily represent the heat flows in the room. This provides data-driven insights into the building and can help to identify optimisation potential for the building. However, the selected features did not adequately map the effect of the thermal mass. In further studies, lagged features or sequence-to-sequence models should be used to map the thermal inertia of the building. Based on this, it should be verified whether one of these methods leads to a better mapping of the heat flows by the local SHAP values.

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